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PYTHON PROJECT REPORT -
CAFÉ MANAGEMENT SYSTEM

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ABSTRACT

The Café Management System is a Python-based application developed to simplify and automate the basic operations of a café. The system provides a graphical user interface where users can view the café menu, enter the quantity of food and beverage items, and automatically generate a bill. The program calculates the total cost based on the selected items and displays a formatted receipt with the current date and time. The system also stores order details in a text file for record keeping and future reference. This project is implemented using Python with the Tkinter library for GUI development and the datetime module for generating timestamps. The application demonstrates practical implementation of programming concepts such as functions, loops, dictionaries, and file handling. Overall, the system offers a simple and efficient solution for managing café orders digitally.

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DESCRIPTION:

In many small cafés and food outlets, order management and billing are often handled manually using paper or basic calculators. This manual process can lead to several problems such as calculation errors, slow billing, difficulty in maintaining order records, and inefficient management of daily transactions.

Manual billing also makes it difficult to track past orders and analyze business operations. As the number of customers increases, managing orders manually becomes time-consuming and inefficient.

To address these issues, a computerized system is required that can automate the process of taking orders, calculating bills, and storing order records. The Café Management System provides a simple digital solution where users can enter item quantities, generate bills automatically, and store order details. This improves accuracy, speeds up billing, and helps maintain organized records of transactions.

1. TOOLS AND LIBRARIES USED

1. Python

Python is the programming language used to develop the entire application. It provides simple syntax and powerful libraries for building applications.

2. Tkinter

Tkinter is the standard GUI library in Python used to create graphical user interfaces. It is used to design the application window, buttons, labels, input fields, and receipt display.

Purpose:

- Creating application window
- Designing menu layout
- Accepting user input
- Displaying bill receipt

Example used in program:

```
import tkinter as tk
```

3. Datetime Module

The datetime module is used to obtain the current date and time when a bill is generated.

Purpose:

- Print invoice date
- Record transaction time

Example:

```
from datetime import datetime
```

4. File Handling (Text File)

File handling is used to store order details in a text file named **orders.txt**.

Purpose:

- Maintain order records
- Store billing history

Example:

```
with open("orders.txt","a") as file:
```

2. SYSTEM DESIGN

The workflow of the Café Management System is shown below.

Step 1: Start Program

The program launches the graphical user interface using Tkinter.

Step 2: Display Menu

The system displays the café menu with item names and prices.

Example:

Coffee	₹50
Tea	₹30
Sandwich	₹80
Cake	₹60
Burger	₹100

Step 3: User Enters Quantity

The user enters the quantity for each item in the input fields.

Example:

Coffee	2
Tea	1
Cake	1

Step 4: Bill Generation

When the Generate Bill button is clicked:

- The program reads the quantities
- Calculates the cost of each item
- Computes the total bill

Step 5: Display Receipt

A formatted receipt is displayed in the receipt area containing:

- Ordered items
- Quantities

- Individual costs
- Total amount
- Date and time

Step 6: Store Order

The order details are saved in orders.txt for record keeping.

Step 7: Clear / New Order

The user can clear the fields and start a new order.

3. IMPLEMENTATION OF CODE

The system is implemented using Python and Tkinter. The program consists of several components and functions that handle menu display, order input, bill calculation, receipt generation, and order storage.

Menu Storage

Menu items and prices are stored using a Python dictionary.

Example:

```
menu = {
    "Coffee": 50,
    "Tea": 30,
    "Sandwich": 80,
    "Cake": 60,
    "Burger": 100
}
```

A dictionary is used because it allows efficient storage and retrieval of menu items and their corresponding prices.

GUI Interface

The graphical interface is created using Tkinter widgets such as:

- Label (for titles and menu text)
- Entry (for quantity input)
- Button (for actions like Generate Bill and Clear)
- Text (for displaying the receipt)

The interface is divided into two sections:

1. Menu Input Section
2. Receipt Display Section

Bill Calculation

The program reads the quantity entered by the user and calculates the cost using the formula:

Cost = Quantity × Item Price

Example code:

```
cost = qty * price
total += cost
```

The total amount is calculated by summing the costs of all selected items.

Receipt Generation

After calculating the bill, the program prints a formatted receipt in the text area containing:

- Item name
- Quantity
- Cost
- Total bill
- Date and time

Order Storage

Order details are stored in a text file using file handling.

Example saved data:

Date:		10-03-2026		14:10
Coffee	x	2	=	₹100
Tea	x	1	=	₹30
Cake	x	1	=	₹60

Total = ₹190

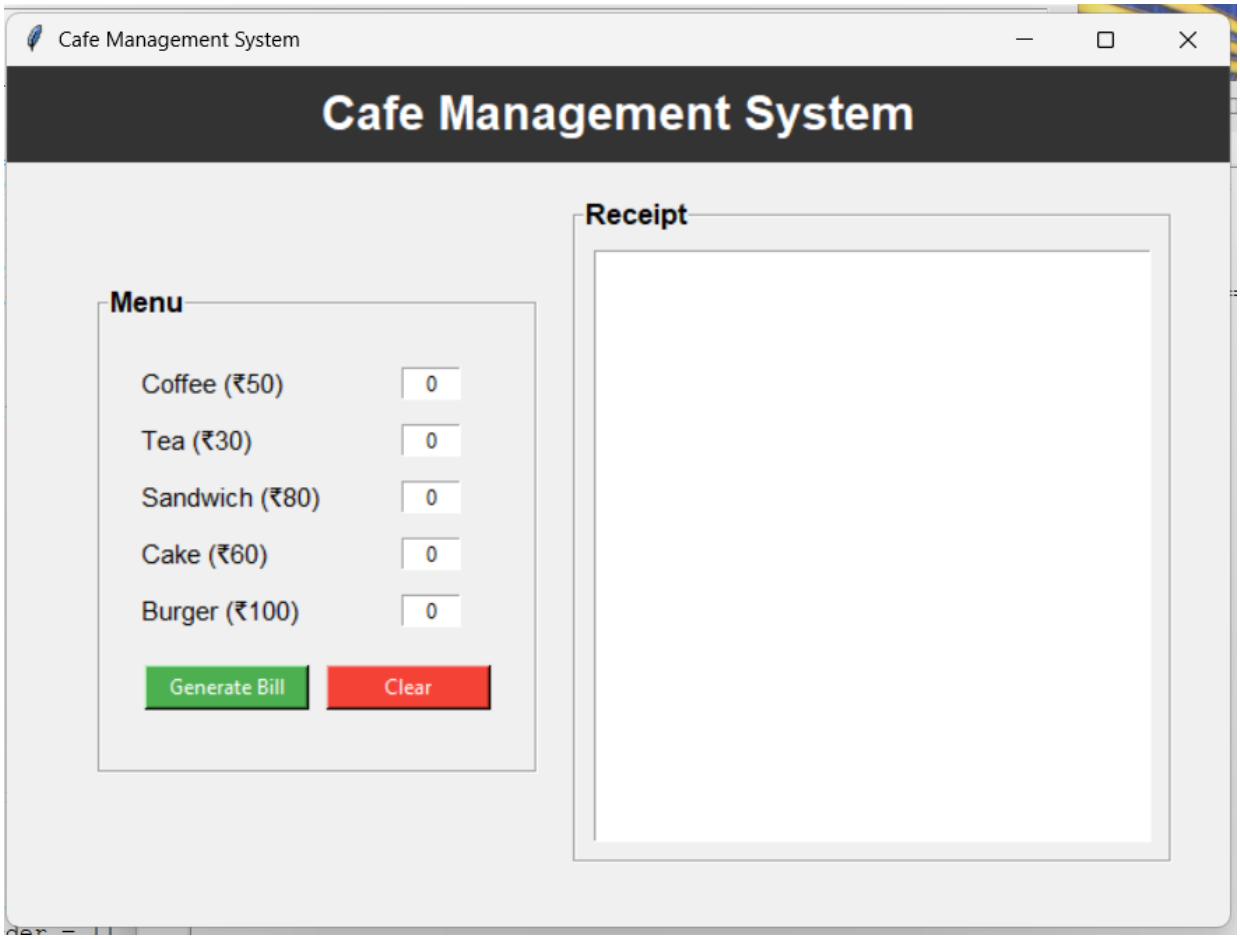
This helps maintain a record of all transactions.

4. RESULTS AND OUTPUT

The application successfully performs the following tasks:

- Displays café menu in the graphical interface
- Accepts user input for item quantities
- Automatically calculates the total bill
- Generates a formatted receipt
- Stores order details in a file

Screenshot 1: Main GUI Interface



Screenshot 2: User Input Example

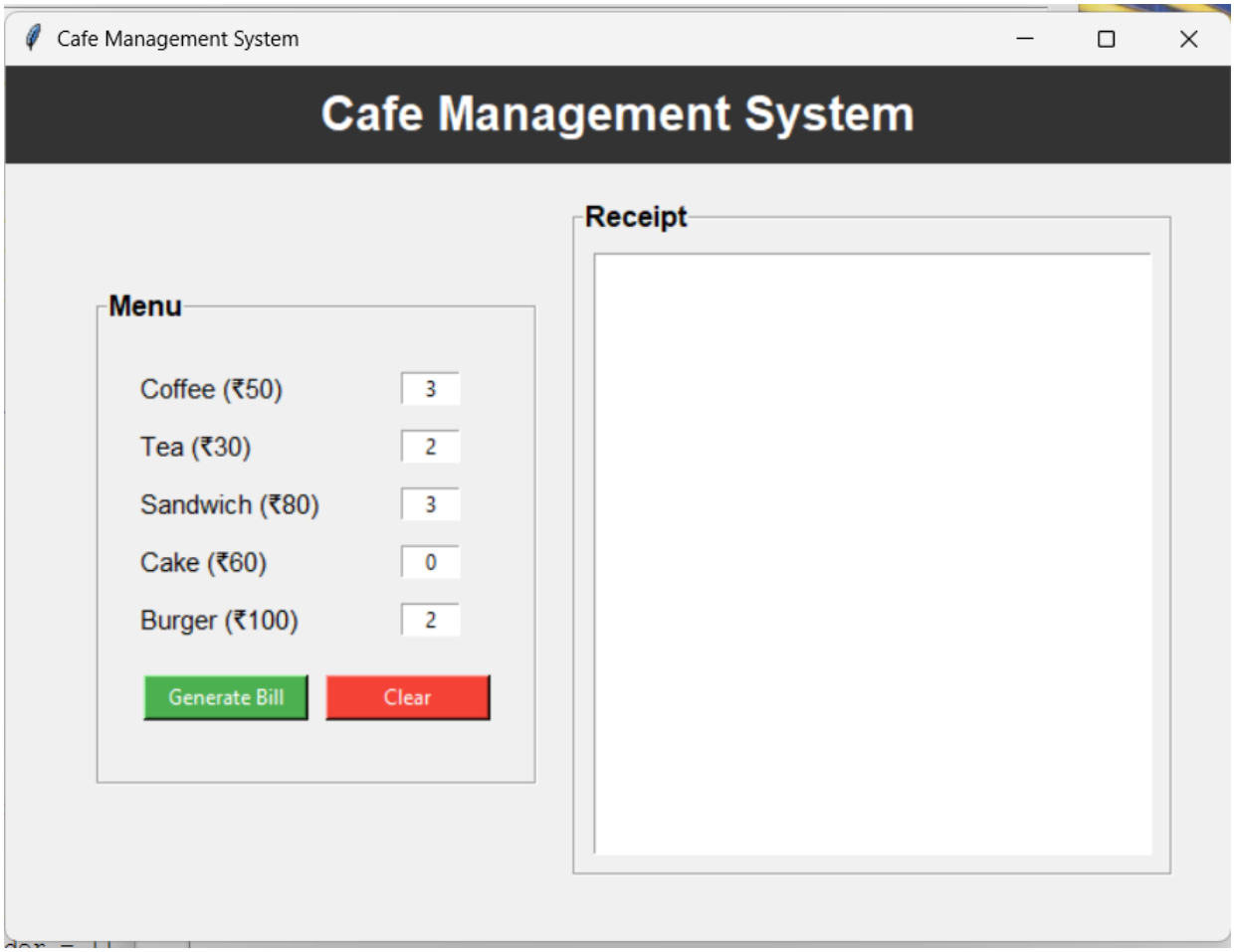
Example input:

Coffee (₹50) = 3

Tea (₹30) = 2

Sandwich (₹80) = 3

Burger (₹100) = 2



Screenshot 3: Generated Bill / Receipt

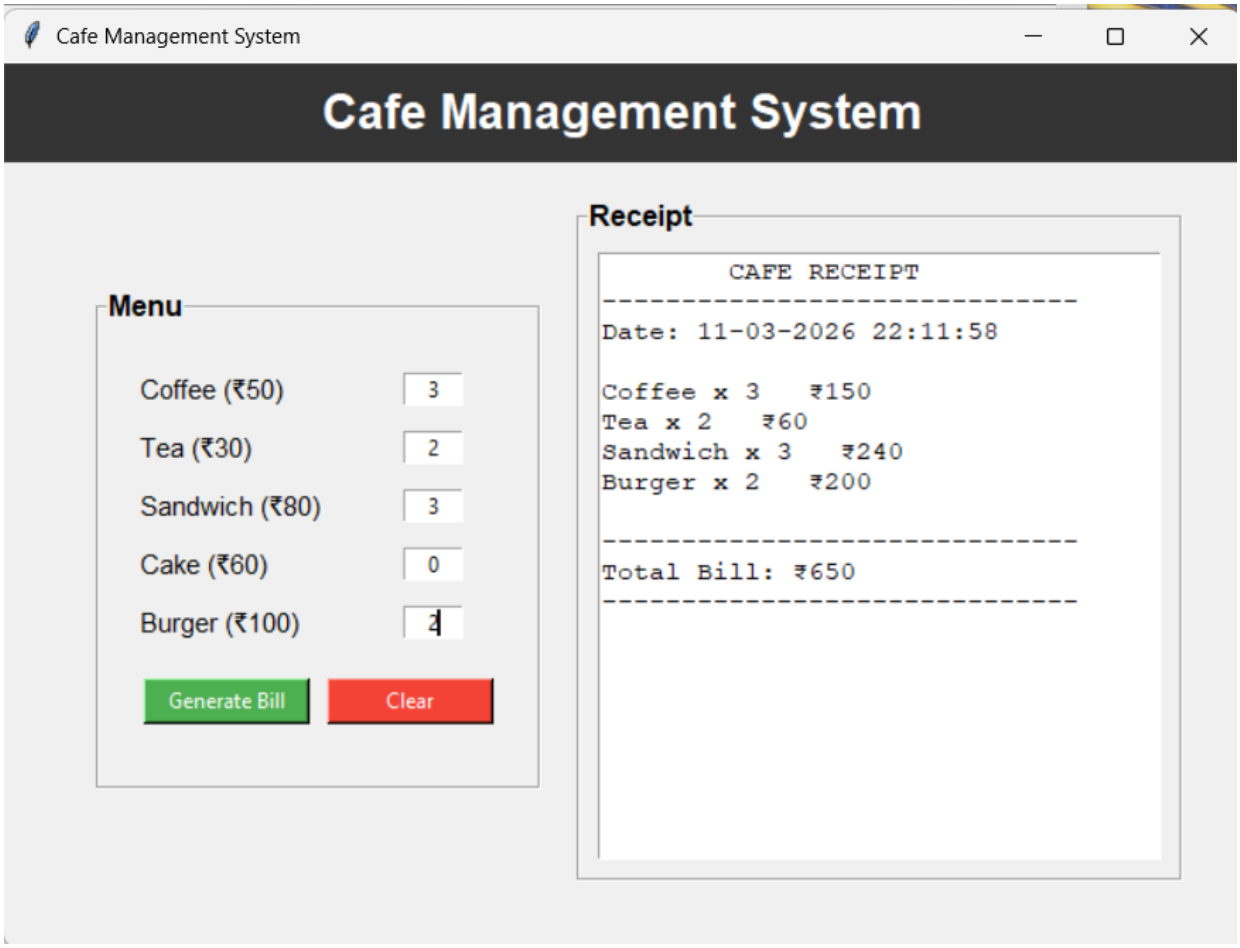
Example output:

Coffee x 3 = ₹150

Tea x 2 = ₹60

Sandwich x 3 = ₹240

Burger x 2 = ₹200



5. Conclusion & Scope for Mini Project Extension

The Café Management System successfully demonstrates the practical implementation of Python programming concepts in a real-world application. The system simplifies the process of order management and billing in a café by providing a graphical interface where users can enter item quantities and generate bills automatically. The use of file handling ensures that order records are stored for future reference.

The project also highlights the usefulness of Python libraries such as Tkinter for building simple desktop applications. By automating the billing process, the system improves efficiency, reduces calculation errors, and maintains organized transaction records.

Scope for Future Enhancement

The project can be further improved by adding the following features:

- Integration with a database such as MySQL or SQLite
- Staff login authentication system
- Daily sales report and analytics
- Graphical dashboard for revenue tracking
- Online payment integration
- Menu images and advanced GUI design

These enhancements can transform the system into a fully functional café management application.